

From Waste to Watts: Predictive Modelling and Environmental Insights from the US Livestock Farms

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Declaration

I, **Zahra Qamar**, declare that the work in this dissertation titled “*Waste to Watts: Predictive Modelling and Environmental Insights from US Livestock Farms*” is carried out by me. This work has not been submitted to Anglia Ruskin University or any other educational institution for the award of a degree or educational qualification. I also declare that the information published in this dissertation has been obtained and presented in accordance with academic rules and ethical conduct. Any information obtained from other sources has been properly referenced.

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Abstract

Anaerobic digesters have emerged as a key component of sustainable agriculture due to the increasing need for renewable energy sources and the need to lower greenhouse gas emissions. Digester adoption in the US is still restricted because of regional and livestock specific variability, high costs and unpredictable results. This research highlights the gaps in accuracy, interpretability and regional analysis by evaluating and predicting the performance of livestock based anaerobic digesters. The dataset "Livestock Anaerobic Database" comprised of 491 projects from 1979 to 2019. After preprocessing and some initial steps, Random Forest classifier and XGBoost regressor were trained to predict two important features: Biogas Generation and CO₂ emission reductions. Mean Absolute Error (MAE) and R² were used to evaluate the model's performance. Additionally, SHAP analysis proved itself useful in improving the interpretability of the model. Incorporating a geospatial extension of the dataset made it possible to map regional digester adoption patterns. Based on the results, Random Forest and XGBoost both outperformed conventional regression techniques and attained high prediction accuracy of R² > 0.9. Although swine based digesters have shown more variability and poultry ones showed effectiveness even though their herd sizes were really small as compared to the two, dairy based digesters dominated in terms of scale. Regional analysis revealed that digester adoption is strong in California, Wisconsin and New York. Shapley Additive Explanations (SHAP) values identified livestock numbers, waste volume and efficiency of gas generation per animal as the strongest predictors. Livestock numbers directly influence the waste and methane potential, waste volume describes the feedstock that enters the system, similarly, generation of gas generation per animal directly affects emission reductions. This research contributes to machine learning driven, multi-livestock and region specific analysis of US digesters. By combining interpretability, spatial analysis and predictive modelling, it provides information to farmers, stakeholders and policymakers. The results illustrate that anaerobic digesters can reduce emissions and produce renewable energy but their effectiveness is dependent on regional conditions, livestock traits and other regulations.

Chapter 1

Introduction

1.1 Overview

The world's energy market is experiencing a significant shift as the countries are struggling with the issues of climate change and increasing energy consumption. Rapid growth in population and industrialization has impacted the greenhouse gas (GHG) emissions and air pollution in an unpleasant way and has also caused the energy demand to spike by over 150% over a span of 40 years. Conventional methods rely heavily on fossil fuels, which is a cause of rise in greenhouse gas emissions. There is no denying the fact that approximately 80% of the world's energy comes from fossil fuels, but because their supply is limited and the effects are highly negative, an alternative is required [Lorin et al., 2019]. Greenhouse gases, particularly Carbon dioxide (CO₂) have a significant impact on global warming. This makes renewable energy solutions a necessity. That is where biogas production through anaerobic digestion of livestock manure comes in, as it not only provides for the generation of renewable energy but also provides sustainable waste management plans. Anaerobic digestion (AD) is a multistep biological process in which a group of microbes decomposes organic matter without oxygen. It includes hydrolysis, acidogenesis, acetogenesis and as illustrated in Figure 1.1 [Cowley and Brorsen, 2018, Hilkiyah Igoni et al., 2008, Korres et al., 2013]. This process of anaerobic digestion yields biogas, which is usually made up of trace gases, 30-45% CO₂ and 55-75% methane.

In the United States, livestock farms are using anaerobic digesters more and more. These digesters are utilized for converting animal waste into biogas, which is a mixture of methane and CO₂. Biogas is important for electricity generation, grid injection and transportation in the form of bio methane. It also has a calorific value of 4500-5000 kcal/m³ which helps replace fossil fuels [Angelidaki et al., 2018]. These anaerobic digesters are essential for environmental sustainability as they lower the quantity of methane emissions that would otherwise leak and destroy the atmosphere. Because of their high cost, unpredictable returns and inconsistent performance, digesters are still only used by 2% of

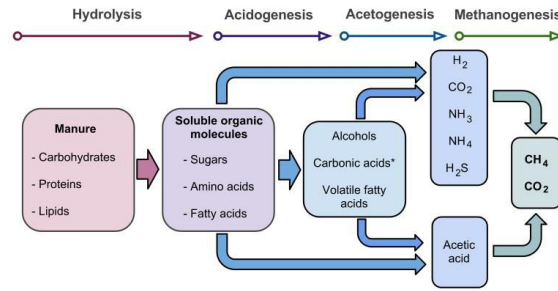


Figure 1.1: Process of Anaerobic Digestion. Source: [Nour and Nour, 2017]

licensed livestock farms [Cowley and Brorsen, 2018]. It is quite difficult to determine the actual advantages of the digesters because of wide variations in the digester types, livestock groups and the methods of how the digesters are operated [Hilkiah Igoni et al., 2008].

Livestock manure is an unexplored energy source and an environmental problem in agricultural areas. Livestock farming accounts for around 14.5% of global GHG emissions, with manure management significantly increasing the release of methane and nitrous oxide [Tryhuba et al., 2025]. In Turkey, rules implemented in 2021, encourage farms to store and use manure for biogas generation, where there seems to be unrealized potential for biogas generation from cattle, small ruminants and poultry [Pence et al., 2024]. Utilizing this manure for the production of biogas might significantly decrease the emissions and help meet national energy goals. This highlights the strategic importance of biogas in local energy planning and climate action w.r.t the geographic locations.

Therefore, anaerobic digester (AD) is waste-to-resource pathway that addresses social, economic and environmental aspects of sustainability in addition to being an energy solution [Mata-Alvarez et al., 2014]. Still, process instability, feedstock unpredictability, scalability, industrial facilities and high operating expenses are some of the obstacles that AD must overcome to achieve its full potential.

It is important to comprehend the performance, efficiency and potential for energy generation of digesters across various livestock types for directing investment and policy decisions. Although earlier studies have frequently concentrated only on dairy farms [Cowley and Brorsen, 2018], there aren't many analyses involving a variety of animal species. Moreover, with advancements in machine learning (ML), analysis of extensive datasets, energy production forecasting, calculation of emission reductions and evaluation of digester efficiency can be performed easily [Sappl et al., 2023].

In order to fill these gaps, this study applies machine learning models to the US Livestock Anaerobic Digester Database, which comprises of approximately 491 projects spread across several states. This research helps in identifying efficient systems for renewable energy and ecological sustainability by forecasting electricity generation, CO₂ emission

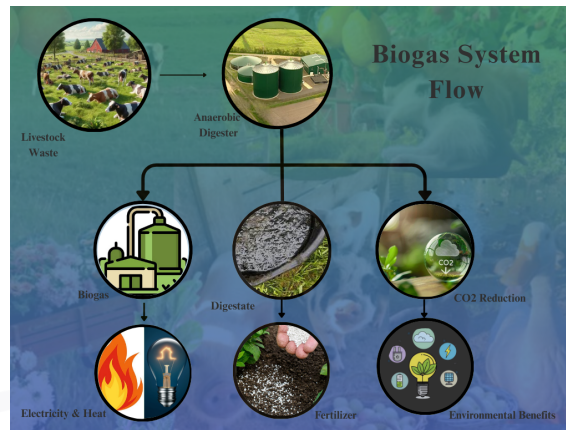


Figure 1.2: Biogas System Flow

reduction and comparing performance across livestock types and regions.

1.2 Problem Background

The installation of ADs on livestock farms is a way to reduce GHG emissions and produce renewable energy. Generally, AD systems have two advantages: preventing methane from being emitted into the atmosphere and turning animal manure into biogas which is further used for heat and energy [Hilkiah Igoni et al., 2008, Weiland, 2010]. This flow is depicted in Figure 1.2. Even with this, there are still a number of technical, financial and policy related obstacles that prevent the widespread use and performance of livestock digesters. Digesters are utilized in providing useful co-products such as bedding and fertilizer while lowering methane emissions, odors and nutrient runoff. In US, however, out of 20,000 concentrated animal feeding operation (CAFO), only 238 adopted digesters, that makes it about 1% only [Cowley and Brorsen, 2018]. The limited adoption of these digesters is because of variable energy yields, high capital costs and uncertain profitability. There is a need to learn more about the most effective digester designs, feedstock and farm sizes as well as how these affect energy production and emission reductions to highlight the reasons for this low adoption. Some interconnected variables such as the substrate type, temperature, pH and microbial dynamics also influence the effectiveness of the AD systems [Hilkiah Igoni et al., 2008, Weiland, 2010]. Other than these variables, feed and animal management, high moisture content and different nutrient compositions are some more issues that are associated with livestock manure. Variability in the performance of digesters is a persistent issue.

Due to the variations in solid content, nitrogen levels and biodegradability, digesters that are made for dairy manure could not function well with swine or poultry waste. Although the study [Hilkiah Igoni et al., 2008] stressed that the digester design needs to be tailored to the unique features of local waste but in reality, a one-size-fits-all design is frequently

used which lowers the efficiency and raises the risk of failure.

Scalability is another technological constraint. Scaling up to industrial operations causes operational uncertainty that is difficult to capture in controlled settings even if laboratory research sometimes give encouraging results [Li et al., 2022]. Another major obstacle to adoption is the economy. Digesters have significant fixed costs and only larger farms with economies of scale can be profitable. Revenues are unpredictable and needs incentives frequently to compete with traditional energy.

Methane and nitrous oxide emissions and odor problems are caused by improper manure management. Digesters can drastically lower animal farms' carbon impact by absorbing methane. Some benefits include better cleanliness, odor management and nutrient recycling [Weiland, 2010].

Apart from this, it is difficult to predict AD's energy output and emission savings. Mechanistic models such as the Anaerobic Digestion Model No. 1 (ADM1) can be used to stimulate biochemical processes but they require a lot of input data which are rarely available at industrial farms [Li et al., 2022, Batstone et al., 2002]. These models are therefore, not well suited for making real time decisions. Statistical models fall short in capturing the dynamic and nonlinear character of anaerobic digestion process. They have limited predictive potential, particularly when dealing with farm-level data.

Utilizing ML techniques such as Random Forests, Gradient Boosting and Neural Networks, an accurate estimation of methane yield and instabilities can be identified even if the input is incomplete or noisy [Sappl et al., 2023]. When ML algorithms are applied to the digester data from the US, the algorithms perform better than linear regression models [Hassan et al., 2025]. Moreover, in Ukraine's cattle farms, neural networks play an essential role in achieving a balance between energy recovery and emission reduction [Tryhuba et al., 2025]. Not only that, the Brazilian biogas plants show the scalability benefits of using ML models [Balakrishnan et al., 2025]. These examples while demonstrate that there is an increase in the usage of ML in this field, it can not be denied that the livestock digesters in the US are not studied well enough.

There aren't many digester performance comparisons between different animal types in the US contexts because the majority of studies concentrate almost entirely on dairy systems [Cowley and Brorsen, 2018]. Moreover, the defects in policy and the economy are apparent. A few of the studies have examined the profitability of various digester designs, livestock scales and the impact of regional energy policies on adoption but the results have not been sufficiently assessed [Cowley and Brorsen, 2018, Pence et al., 2024]. A number of gaps still exist in modelling methodologies. For example, ADM1s are data intensive and unfeasible [Balakrishnan et al., 2025] for industrial scale applications. Despite providing promising evidence of their predictive ability. ML applications are still under used in livestock digesters [Li et al., 2022, Hassan et al., 2025]. Lastly, there aren't

many comparisons of CO₂ reduction across various livestock types [Cowley and Brorsen, 2018]. Furthermore, regionally focused strategies used in countries like Turkey, needs to be implemented within the US as well because mapping of spatial variance is necessary [Pence et al., 2024].

This study focuses on the US livestock digesters, predicts energy generation and CO₂ emission reductions using ML approaches. It also tackles the long-standing bias in the existing studies towards dairy systems by comparing performance across livestock categories such as swine and poultry [Cowley and Brorsen, 2018]. Additionally, the study uses geographical analysis, providing a fresh perspective on regional patterns and future possible hubs for the growth of biogas in the US [Pence et al., 2024]. This would support the development of data driven strategies for the US agriculture sector to produce renewable energy and cut emissions.

1.3 Research Aim

The aim of this study is to predict energy generation and CO₂ emission reductions from animal waste digesters in the US using machine learning algorithms. The purpose is to identify the best livestock based biogas projects for climate mitigation and the production of renewable energy. This is to be done by comparing the performance of the digesters across various livestock types and by examining regional differences.

1.4 Research Objectives

- To predict biogas energy generation and CO₂ emission reduction from US livestock farms using machine learning algorithms and statistical techniques.
- To compare efficiency and performance of digesters across different livestock types.
- To employ advanced visualization frameworks in Python for the prediction analysis and outcomes.
- To use spatial mapping and explore regional patterns for potential future operation hubs for anaerobic digesters.

1.5 Research Scope

The scope of this study is defined around the use of publicly accessible US Livestock Anaerobic Digester Database from Kaggle which includes the information on 491 projects in several states. Some of features are animal type, digester type, waste generated,

biogas produced, electricity output and estimate annual CO₂ reductions. The focus is limited to livestock based anaerobic digesters and does not include codigestion with other feed stocks and non agricultural biogas facilities[Pence et al., 2024] . Since electricity generation (kWh) and CO₂ emission reductions are the most important measures of energy efficiency and climate benefits, their results are the ones that are taken into account in the study [Weiland, 2010]. The study's geographic scope is limited to the United States and state level regional comparisons are made. Broader worldwide comparisons are only presented as contextual references based on the livestock biogas research being carried out in different countries [Pence et al., 2023]. The dataset's temporal scope is projects from 1979 to 2019, hence the study offers a historical and data driven assessment of digester efficiency within the time frame.

1.6 Abridged Methodology

The US Livestock Anaerobic Digester Database records about 491 anaerobic digester projects in several states and serves as the focal point of the study's data driven technique. Pandas and NumPy are used in Python for data preparation and label encoding is used to numerically encode the categorical variables. Project Age is a derived variable that is included to account for any kind of variations in digester performance between older and newer models.

ML models are developed to predict these two significant outcomes: biogas generation and CO₂ emission reductions. These two are the essential markers of both energy efficiency and climate benefits [Li et al., 2022, Weiland, 2010]. Because it can manage non-linear correlations and interactions between input features, a Random Forest classifier alongwith XGBoost regressor are the main predictive models used in the study. The model outputs are evaluated using statistical metrics such as the coefficient of determination (R^2) and Mean Absolute Error (MAE), which quantifies predictive accuracy and deviation from the observed values [Sappl et al., 2023]. To evaluate the digester performance, the dataset is split into training and test sets (80:20). Because of their better performance in earlier research, additional ensemble boosting methods are also used to assess the robustness of the research [Hassan et al., 2025]. Interpretability was the highlight in this study. SHAP is used to provide a more thorough understanding of the behavior of the models, while Matplotlib, Seaborn and Plotly are used to generate visuals. Finally, geographic analysis is used to investigate regional variations in digester performance using mapping tools like Folium and GeoPandas. In order to balance predicted accuracy, interpretability and geographic context, this study offers a methodology for evaluating livestock digesters using machine learning.

1.7 Contribution

- The study fills a gap in previous research by offering the first machine learning (ML) driven comparative analysis of US livestock digesters across various livestock kinds. [Cowley and Brorsen, 2018, Hassan et al., 2025].
- The research study develops a predictive framework for estimating the generation of electricity and the reduction of CO₂ emissions, the two outcomes crucial to the policy of renewable energy [Li et al., 2022].
- The study incorporates a geographical component in it.
- The analysis demonstrates the superiority of ML methods over conventional regression models [Hassan et al., 2025, Tryhuba et al., 2025].
- Lastly, the report generates insights that can be included in the policies and can be practiced. This could help the regulators, investors and farmers scale livestock digesters as a sustainable energy mix in the US.

Chapter 2

Literature Review

2.1 Overview

Biogas is an important renewable energy resource mainly in livestock farms, where they are not only used as an energy source but also helps reduce environmental problems. Animal manure from these farms can be converted into a usable form of energy by passing the manure from biogas digesters [Hilkiah Igoni et al., 2008, Cowley and Brorsen, 2018]. This results in the generation of a mixture consisting of carbon dioxide and methane. The process helps in the reduction of greenhouse gas emissions that are harmful for the environment. Several studies confirm that livestock manure can be effectively used to generate electricity, but its widespread adoption was difficult due to the financial and technical constraints [Kemausuor et al., 2014]. With an increase in technological practices, understanding biogas digester operations has become easier, thus, more contribution is expected in optimizing biogas system operations and implementing sustainable agricultural techniques.

Majority of the studies emphasize the advantages of AD for the environment but they overlook the operational difficulties that are associated with it. Digester failures, maintenance expenses, regional limitations are a few examples that come under the heading of operational difficulties [Weiland, 2010].

2.2 Understanding Biogas Digesters

Biogas digesters are industrial machines that take up the livestock waste and converts it into renewable energy through anaerobic digestion. Anaerobic digestion is carried out when an oxygen deprived environment is created in the digester, microorganisms break down manure and produce biogas which is a mixture of methane and CO₂. Usage of these anaerobic digesters in livestock farms reduces water and air contamination thus helps in

controlling environmental pollution [Tshemese et al., 2023]. Similarly, anaerobic digesters provide us with by-products such as nitrogen and phosphorus and also increases soil fertility [Alengebawy et al., 2024]. Moreover, rise in energy generation, waste management, reduction in CO₂ emissions are some of the advantages for using biogas digesters.

Anaerobic digestion process consists of the following stages: hydrolysis, acidogenesis, acetogenesis and methanogenesis. There are different types of digesters such as the covered lagoons, complete mix, plug flow, fixed film and many others. All these have the same basic principle that is the breakdown of organic matter utilizing microorganisms and in the absence of oxygen, though they differ in configuration and temperature. The digesters are employed in farms based on the requirements and efficiency.

Livestock such as dairy, swine and poultry are majorly involved in biogas production projects because they contribute with different types of organic waste for anaerobic digestion [Alengebawy et al., 2024]. Large scale biogas systems use dairy manure because it is high in fiber and so is highly effective. Since swine manure is high in nitrogen, it is suitable for projects where large amounts of methane production is expected. Poultry waste is also rich in nutrients and can enhance the digestion process [Weiland, 2010].

Some of the digesters that use wet fermentation systems and have demonstrated great performance in terms of yield and overall efficiency. These systems are able to handle high moisture content such as the cattle and swine manure and offer optimal conditions for microbial activity [Weiland, 2010]. Along with that, the feedstock has to be adaptable to the digester type and so, the type of feedstock selected go hand in hand with the type of digester being used and affects the performance as well [Abubakar, 2022]. Table 2.1 displays various types of digesters, their feedstock, applications and limitations.

Table 2.1: Comparison of Different Digester Types. Source: [Alrowais et al., 2023]

Digester Type	Feedstock (%)	Applications	Advantages	Limitations
Plug Flow	11-13 (Ruminant Manure)	Dairy farms	Simple design	Requires consistent waste input
Complete Mix	Less than 11 (Slurry)	Mixed farms, Wastewater	Can handle variable wastes	Higher energy input
Covered Lagoon	Less than 2 (Liquid)	Warm Climates, Swine Waste	Low cost	Low gas yield, Temperature sensitive
High Solids	Greater than 15 (Crop Residues)	Municipal Solid Waste	Can handle solid-rich waste	Costly, Complex design

Furthermore, because it influences biogas production and system performance, the digester's configuration must match the operating circumstances and waste profiles. Moreover, environmental factors like humidity and temperature have an effect on microbial activity, which in turn affects digester performance.

In addition to lowering the increase in waste problems, the installation and operation of anaerobic digesters is crucial since it provides a sustainable energy source. [Tshemese et al., 2023]. By reusing agricultural byproducts and livestock manure, the anaerobic digestion process lowers the total waste footprint of livestock farms. In the agricultural industry, combining waste reduction and energy production would help increase sustainability within the sector.

2.3 Environmental Impact of Biogas

Producing biogas on animal farms is essential for lowering greenhouse gas emissions into the atmosphere and instead absorbing and using them. The production of biogas significantly lowers global warming caused by the powerful gases: CO₂ and methane. The production of biogas from animal and slaughterhouse waste reduced CO₂ emissions by around 3.631 billion kilos per year [Afotey and Sarpong, 2023]. Additionally, the production of biogas from livestock farms emphasizes a critical sustainability aspect and a fossil fuel substitute [Hassan et al., 2025]. Farms can not only create manure into an energy resource but can also decrease their reliance on non-renewable resources.

Other than reducing greenhouse gas emissions, biogas digesters affect ecological systems as well, by boosting the air and soil quality and by minimizing the farms' odors. The odor is mainly because of the release of nitrogenous compounds, but the digestion process can limit its release [Tshemese et al., 2023]. Other than that, soil fertility can be increased with the help of digestate, a by-product of biogas production. Soil fertility can escalate crop yield and promote sustainable agricultural practices. Although, there are concerns related to the over application of digestate which have led to nutrient leaching. Thus, biogas production provides us with several sustainable environmental benefits but it is important for the agricultural management to ensure regular monitoring for preserving the environment.

2.4 Geographical Aspect of Biogas Production

Introducing visuals and geographical mapping enhances the understanding of digester efficiency and performance. Employing tools such as geographic information system (GIS) technologies reveal spatial patterns across different regions [Jeong et al., 2021]. With a combination of ML models and these visualization tools, stake holders can dynamically interact with data and enable real time monitoring of digesters. This not only aids in identifying high performing digesters but also assists in strategic decision making. It was observed that combining satellite data with livestock and farm characteristics, provides with an understanding of concentrations and emission patterns [Bi and Neethirajan, 2024]. Using these techniques can support more effective renewable energy planning.

Biogas production varies according to environmental conditions and agricultural practices as well. For example, the efficiency and output of biogas is influenced by management strategies and agricultural outputs as well [Weiland, 2010]. As seen in Northern Hesse, Germany, increased biogas capacity correlated with a rise in maize cultivation. This shows that the capacity of biogas plants highly affects the land usage. Incorporating satellite methane data with local dairy farm characteristics significantly influences methane

emission strategies, thus optimizes regional biogas production. Customizing biogas systems to align them with specific geographical features enhances not only the economic stability but also the environmental effectiveness of the renewable energy projects.

2.5 Machine Learning & Analysis

Advancements in machine learning has paved the way for improvement in the predictions of energy output and CO₂ emission reduction in biogas projects. Algorithms such as random forest, artificial neural networks and extreme gradient boosting help in the interpretation of complex parameters and effectively predict bio methane production [Weiland, 2010]. These models work on large datasets and identify influential factors like the volatile fatty acids, feed volume and then predict potential energy yield and environmental impacts [Zhang et al., 2014]. Exploring such techniques not only increases the biogas production but also provides us with a deeper understanding of the variables that effect the performance and efficiency of the systems. However, a difference in system design and equipment might impact some calculations and predictions but despite these challenges, the implementation of machine learning for such tasks holds significant potential for the stabilizing systems and optimizing biogas production.

The process of biogas digester related predictions start with the selection of suitable models for the prediction. Identifying models that can handle complex nonlinear interactions. Random forest, artificial neural networks, LightGBM and XGBoost outperform other algorithms in terms of accuracy and interpretability, capturing intricate patterns in data. Choosing an accurate model is essential as it impacts the accuracy of predictions [Hassan et al., 2025].

- **Random Forest:** Offers feature importance rankings, can handle overfitting and works well with non-linear tabular datasets [De Clercq et al., 2020].
- **XGBoost:** Is mainly used for prediction tasks, achieves good accuracy and computation efficiency.
- **Support Vector Machines (SVM):** Are algorithms that work well with small datasets but are sensitive to any changes in parameters.
- **Artificial Neural Networks (ANN):** Can capture intricate correlations but needs a lot of training data.

In particular, XGBoost was able to successfully highlight some of the crucial predictive features among machine learning models like waste efficiency and daily waste input, which are quite necessary for determining the production results [De Clercq et al., 2020]. The

Table 2.2: Evaluation of Machine Learning models. Source: [Chang et al., 2018]

Model	Strengths	Weaknesses	Evaluation
Random Forest	Can handle non linearity	Can overfit on noisy data	R ² : 0.85, MAE: 1500
XGBoost	High accuracy, Fast	Sensitive to parameter tuning	R ² : 0.90, MAE: 1200
ANN	Can capture complex patterns	Less interpretable	R ² : 0.88
SVM	Works well with high dimensional data	Kernel tuning is difficult, Slow on large datasets	Accuracy: 0.82%
LightGBM/CatBoost	Fast, more effective with categorical data	New, Not widely used in biogas research	R ² : 0.91

Table 2.2 indicates the strengths and weaknesses of some ML models. This angle of research and prediction shows how even slight variations in the waste composition impact digester efficiency. Therefore, the predictions can be made more specific and can be tailored according to requirements. Moreover, to evaluate the accuracy and performance, certain evaluation metrics are also employed. By utilizing K-fold cross validation and learning models, predictions are improved, model accuracy is enhanced and biogas production rate grows [Peng et al., 2024]. Furthermore, when evaluating how well machine learning models perform in predictions, evaluation metrics like R² and Mean Absolute Error (MAE) are crucial. The Random Forest model was found to have an R² of 0.9343, showing a very high accuracy of the model. R² measures how well the predicted values match the actual values [Yildirim and Ozkaya, 2023]. On the other hand, MAE assesses the accuracy of the model by calculating the average measure of errors between the actual and predicted values. These metrics evaluate the model's performance and provides critical feedback [Li et al., 2022].

2.6 Feature Selection

A key component of ML based biogas prediction is feature selection. Since the datasets often contain a large number of variables, not all of them contribute equally to the model's predictive performance. Eliminating irrelevant features enhances generalizability, decreases overfitting and reduces the noise in the data as well. Some of the key factors are:

- **Population size and Livestock type:** Different animals create manure with different volatile solids and methane potential.
- **Microbial activity and Gas production:** are affected by pH, temperature and retention period [Weiland, 2010].
- **Project age and Digester type:** Design affects gas yield and older digesters may have efficiency issues.

In order to provide a more comprehensive knowledge of the responsibilities that each feature plays in the model predictions, SHAP has recently been used. For instance, the

plots can show if the effect is non-linear or whether increasing volatile solids consistently result in higher gas production. These insights are essential for farmers and policy makers. Although, it is evident that feature selection enhances prediction, many researches lack this key element.

2.7 Research Issues

Data analysis performed on the livestock anaerobic data, shows the potential of anaerobic digesters, in reducing greenhouse gas emissions through efficient management of livestock waste. Although, it is important to keep in mind that it is only beneficial to the environment if methane leakage is minimized. The life cycle assessment of anaerobic digestion plants reveal that their environmental profile is beneficial than the conventional electricity generation systems. Such advancements not only support sustainable agricultural practices but also depict the alignment of biogas production with global renewable energy objectives [Lamnatou et al., 2019].

That said, it is important to address the challenges and limitations faced when implementing these technologies on a large scale. One of these is the high upfront cost of setting up anaerobic digesters, which prevents small farms from utilizing these technologies [Tshemese et al., 2023].

Second, a number of variables, including feedstock quality, operating circumstances, geographic location and other little details, affect how efficient the biogas digesters are. This indicates that in order to maximize digesters' performance, a comprehensive custom plan is needed.

Thirdly, accidental methane leakage can have harmful effects in the environment. This highlights the need for regular monitoring and management for these systems. Predictive accuracy could increase if machine learning models that are particularly designed to address anaerobic digestion issues are developed. These models could handle the range of system designs and operational situations by implementing real time monitoring.

Additionally, investigating the use of IoT and sensor networks may help in data collection and real time feedback. This might make it possible to manage the systems more precisely, which would ultimately increase the yield.

By absorbing methane and turning it into electricity, biogas offers livestock farms a sustainable energy source and has a major positive ecological impact. In addition to reducing greenhouse gas emissions, biogas supports sustainable farming methods. Data analysis plays a key role in improving the effectiveness of biogas projects by optimizing digester performance and facilitates decision making for future predictions. Thus, using these biogas systems not only support sustainable energy solutions but also is consistent with energy policies. Table 2.3 shows some of the papers used in this research, there strengths

and weaknesses.

Table 2.3: Literature Review Summary

Study	Purpose	Strengths	Weaknesses	Country
Bioenergy Production by Anaerobic Digestion [Rietveld et al., 2013]	To introduce anaerobic digesters as a sustainable method	Provides a comprehensive review of feedstocks & technologies for anaerobic digestion	Lacks updates after 2017	Ireland & UK
Biodigester and Feedstock Type: Characteristic, Selection & Global Biogas Production [Abubakar, 2022]	To present guidelines for feedstock & digester selection	Offers a criteria for selecting feedstock & digester under varying conditions	Does not have much experimental insights, more on theoretical side	Nigeria
Estimation of Biogas Production Potential & GHG Emission Reduction [Ajakaiye and Surungbo, 2021]	To calculate Ghana's potential capacity for the reduction of GHG and biogas production	Utilizes comparative techniques on livestock & slaughterhouse waste data	No experimental insights available	Ghana
Comparing Mesophilic & Thermophilic Anaerobic Co-digestion for Sustainable Biogas Production [Akemisi et al., 2023]	To calculate production, cost function & feasibility of anaerobic digesters	Makes use of nationwide survey across US dairy & swine operations	Economic feasibility is dependent on co-product markets	US
The Role of Anaerobic Digestion & Biogas in Circular Economy [Anton Paganiotto et al., 2018]	To highlight AD's role in economy	Combines technical, environmental & socioeconomic roles of AD	Lacks region specific details	Sweden, Denmark
From Waste to Energy - Anaerobic Digestion Technologies [Joseph Bilezikci et al., 2024]	To introduce advancements in AD feedstocks, processes & efficiency	Highlights global innovation on AD technologies	Lacks some important details	Spain, Greece
Predictive Modeling of Biogas Production Using ML [Bulakcihan et al., 2025]	To optimize biogas efficiency through predictive modelling	Combines various ML models to improve biogas production	Limited data from a single farm	Brazil
An Empirical Analysis on Renewable Energy: Biogas Production Prediction Using ML [Hassan et al., 2025]	To link dairy farms operations with methane emissions using advanced technologies	Incorporates ML with satellite data to assess the emissions	Limited data i.e. from 11 farms	Canada
Mapping Methane with Satellite Data & ML [Bilal and Nourbajran, 2024]	To evaluate ML approaches for the prediction of biogas	Compares multiple ML models with explainable AI	Small dataset	Bangladesh
Modelling Energy and Emissions from Animal Manure [Pence et al., 2021]	To utilize ML models in estimating manure based biogas production & emissions	Outputs include some high accuracy models	Study limited to 3 provinces	Turkey
Predictive Modeling of Biogas Production using ML [Karki et al., 2021]	To develop ML models for the prediction of cow manure biogas production	Utilizes various ML algorithms on real life data	Limited dataset (from 1 region)	Brazil
Neural Network Models for Sustainable Livestock Waste Management [Tayyaba et al., 2025]	To predict & improve waste generation using neural networks	Shows high accuracy in predicting livestock waste and improving emissions	Model is region specific	Ukraine
Biogas Production: Current State and Perspectives [Weiland, 2010]	To analyze biogas production processes, feedstock & future	Provides review of biogas production technologies & applications	Lacks information on technological advancement after 2010	Germany

Chapter 3

Methodology

3.1 Research Design

To predict the effectiveness of anaerobic digesters in the US farms, this research incorporates quantitative and data driven methodology with some statistical techniques, machine learning and other visualization. Comparing digester performance across various livestock types, estimating biogas energy output and reducing CO₂ emissions are the objectives. The primary steps of this study are data collection, processing, cleaning, feature engineering, model training, testing, classification analysis and explainability using SHAP.

In addition to predicting energy and emission outcomes, the analysis provides an understanding of the factors that significantly influence the results. The study also facilitates a comparative evaluation of digester efficiency. Figure 3.1 depicts the workflow of the whole project.

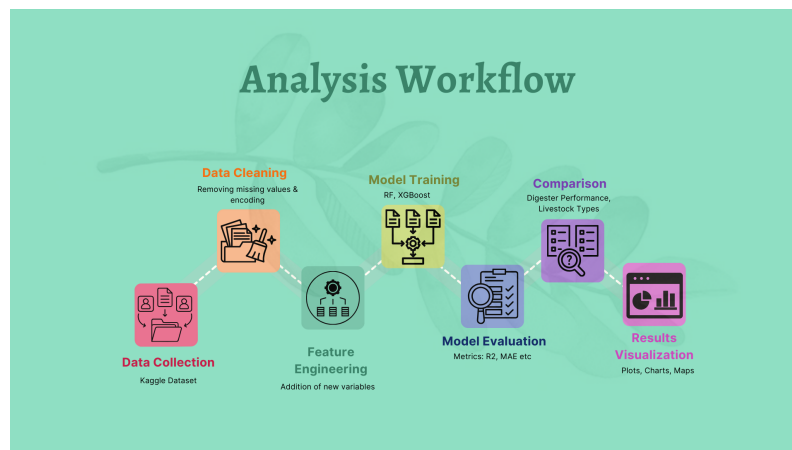


Figure 3.1: General Workflow of the Project

3.2 Data Collection

The dataset being analyzed in this project is sourced from Kaggle and compiled by Mehmet Isik under the heading “Livestock Anaerobic Digester Database” [Isik, 2023]. It includes about 491 projects across the US, with the variables as follows:

Number of animals, Year operational, Livestock type, Digester type, Waste generated, Biogas generation estimate (CU-ft/day), Electricity output (kWh/year), Annual CO₂ emission reductions (kg/year).

The dataset is perfect for both regression and classification tasks for example the prediction of the emission reduction values, biogas outputs and the tasks where digesters are grouped into categories based on the defined thresholds. The dataset holds all the parameters for a comprehensive predictive modelling.

3.2.1 Data Preprocessing

Machine learning models require consistent, numerical data for any kind of analysis, hence, data preprocessing is carried out to make sure that it is ready to use for the study.

3.2.2 Missing values

The digester projects that had missing values for any of the features were deleted.

3.2.3 Categorical Feature Encoding

LabelEncoder, an ML technique, was used to translate the categorical variables into numeric values. Introducing this also helped in the handling of “string” type values in Random Forest and XGBoost [Hilkiah Igoni et al., 2008].

3.2.4 Feature Engineering

A new feature, Project Age, is derived from the existing feature.

$$\text{Project Age} = 2025 - \text{Year Operational}$$

This was necessary because it shows how long a digester has been operating, which can be helpful in calculating the effectiveness as well [Weiland, 2010].

3.3 Machine Learning Models Selection

The main algorithms used in this study are Random Forest (RF) and eXtreme Gradient Boosting (XGBoost). These models have shown capacity to manage complex, non-linear and high dimensional characteristics of biogas datasets. They use bagging (model training

on random samples) and boosting (model training in a way that new model learns from the previous model's mistakes) approach to reduce overfitting and capture complex feature interactions. Overfitting is when a model learns too much from the training data, including the unnecessary information. This results in, the model working perfectly for training data but not when tested on new data [GeeksforGeeks, 2022]. The model starts picking up random fluctuations rather than the patterns. Overfitting can happen because of various reasons i.e. the model being too complex, insufficient training data, outliers in data and other such factors [Aurora, 2024].

RF has been applied in various environmental and agricultural modeling predictions and has proven itself highly useful. One of its ability to determine feature importance greatly improves interpretability in biological and energy related situations. RF has outperformed many statistical techniques in both lab and industrial digestion systems [Chen and Guestrin, 2016].

Likewise, XGBoost achieves greater prediction accuracy in tasks related to renewable energy. It is especially useful for diverse and unbalanced datasets, which are common in biogas production studies. Recent comparative studies verify that XGBoost and some other related boosting techniques perform better than traditional regressors, particularly in biogas yield prediction [Isenkul et al., 2025].

3.3.1 Regression Models

XGBoost Regressor surpasses all the regressors in terms of predictive accuracy and is tuned for speed and performance on tabular data.

Metrics for Evaluation

The coefficient of determination (R_2) tells you how well the independent variables explains the variation in the dependent variable in a statistical model. The average absolute difference between the predicted values and the actual target values is represented by Mean Absolute Error (MAE) [Ahmed, 2023]. The combination of these measurements provide both practical interpretability and statistical validity.

3.3.2 Classification Models

Biogas generation and emission outputs were categorized into “High” and “Low” classes based on the threshold that was 24,100 for biogas and 345000 for CO₂ emission reduction. This was to compare the performance of digesters across livestock farms.

RF classifiers were mainly used along with the classification report and confusion matrices. The classification report provides with precision value, recall score, F1 score and accuracy score. Precision value is the accuracy of the positive predictions, recall score value is the model's ability to find all positive instances and F1 is the mean of these two

values Arora [2024]. Whereas, the confusion matrix shows how many predictions were right and wrong for each class.

By using these ensemble methods, this study achieves high predictive accuracy. Their strong performance in previous anaerobic digestion and livestock emission studies indicates that these models are the best option for modeling biogas production under varying feedstock and operational conditions.

3.4 Model Analysis and Visualization

There is a “black box” theory related to machine learning, Blackbox ML describes models that provide outcomes without disclosing the decision making process [SEON, 2023]. SHAP values are used to address this, as they allocate a contribution value to each feature corresponding to the model’s prediction. SHAP provides information on the features that have greatest impact on biogas production and CO₂ emission reduction, impact of certain features on the efficiency of the digesters and other such comparisons. This sections helps to ensure that the predictions are accurate and understable by improving the interpretability [Lundberg, 2020].

The following statistical models and visualization methods are applied to support the results from the machine learning models:

- **Correlation Matrix:** Displayed linear association between features and outcomes.
- **Box Plots:** Performance distribution comparisons are made across different livestock types [Story, 2020]
- **Geospatial Visuaization:** Mapping digester performance geographically using GeoPandas or Foilum.

Policymakers and stakeholders can understand the results through these visualizations which can be easily interpreted by technical and non-technical audience.

Chapter 4

Results and Discussion

4.1 Overview

The Livestock Anaerobic Digester Database is a dataset of 491 digester projects across the US, from the year 1979 to 2019 [Isik, 2023]. The first step was the preprocessing of data which included some cleaning and initial analysis. A new feature “Project Age” was derived and the following distributions were noted.

Livestock: Dairy farm digesters dominated followed by swine and poultry. As illustrated in Figure 4.1

Digester Types: Covered Lagoon, Complete Mix and Mixed Plug Flow digesters were the most popular ones. As shown as figure 4.2

Electricity Generation: The distribution is highly skewed and shows that digesters produce a fair amount of electricity which is sufficient for the use of a farm. A very small number of digesters produce massive amounts of power. The histogram is quite misleading and so log scaling is included to get a better visualization. Log transformation shows that about 40-60 digesters produce about 10 trillion-1 quadrillion kWh electricity, while most of them just produce 1 - 10kWh. A few outliers depict a huge amount of electricity being generated of about tens of billions of kWh/year. The distribution is depicted in the Figure 4.3.

CO₂ Emission Reduction: CO₂ emission reduction is checked w.r.t the digester types as well as the operational years of the projects. Covered Lagoon and Mixed Plug Flow digesters are the biggest contributors to CO₂ emission reductions on average. As seen in the Figure 4.4 projects with early operational years (new ones) have high variability with few providing massive CO₂ reduction. Whereas, the older ones have smaller reductions. This means that the new projects push towards CO₂ emission reduction. The superiority of dairy based digesters is in line with earlier studies that highlighted dairy manure as the primary feedstock. However, the presence of digesters for swine and poultry, even if in small quantities, indicate a growing diversity. The highly skewed energy generation

Digesters Distribution by Livestock Type

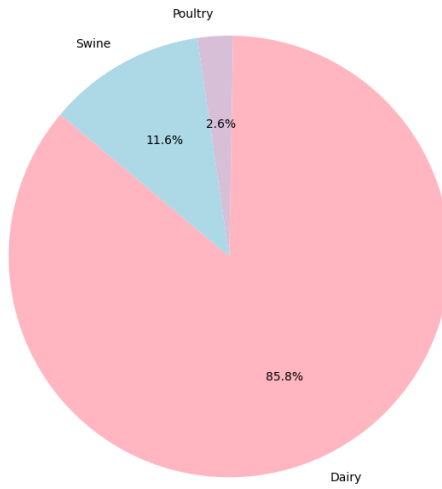


Figure 4.1: Digesters Distribution Based on Livestock

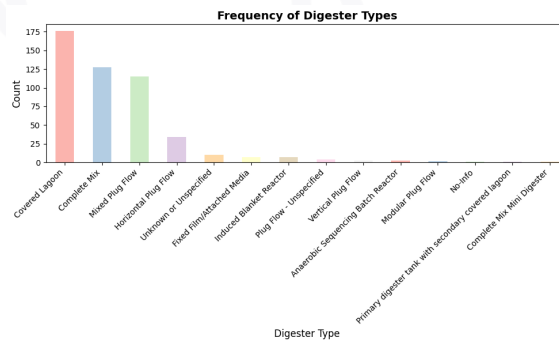


Figure 4.2: Frequency of Digester Types

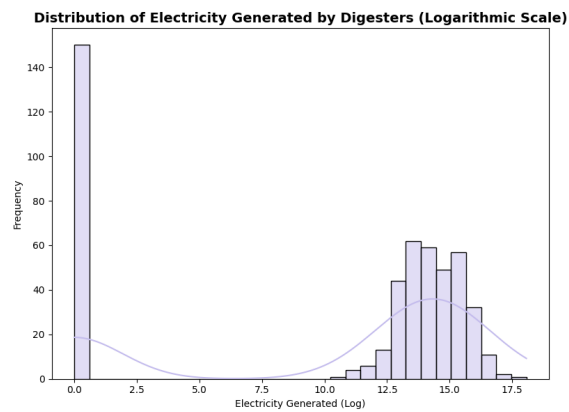


Figure 4.3: Electricity Generation by Digesters (Logarithmic Scale)

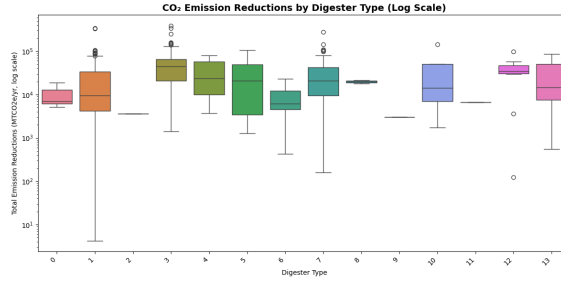


Figure 4.4: CO₂ Emission Reduction by Digester Type

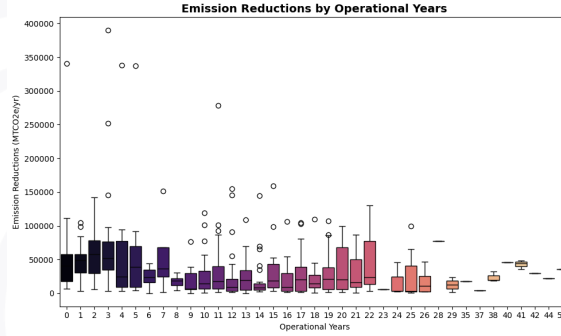


Figure 4.5: CO₂ Emission Reduction Based the Years in Operation

distribution indicates that majority of the digesters produce at smaller scales and some at industrial levels. The broad range of CO₂ emission reduction emphasizes the need for predictive modelling.

4.2 Energy Generation and CO₂ Reduction

XGBoost regressor and Random Forest classifier were utilized for predictive modelling and the results are as follows:

CO₂ Emission Reduction

XGBoost model obtained of a value 0.914 for R^2 and 4945.5 for MAE. This value was achieved after a K-Fold evaluation. By using Random Forest classifier, the values achieved for precision, recall and f1 were extraordinary. The high being 0.95, 0.96 and 0.96 respectively. Additionally, for better understanding of the model, a confusion matrix is displayed in the Figure 4.6.

Electricity Generation

XGBoost model achieved a value of 0.75 for R^2 and 102505.5 for MAE. Random Forest classifier predicted remarkable values for precision, recall and f1. They were 0.98, 0.94 and 0.96 respectively. The confusion matrix also helped in understanding the model, illustrated in Figure 4.7. The outputs from XGBoost confirms the effectiveness of ML models in managing non-linear, multi- dimensional data. Although the accuracy was

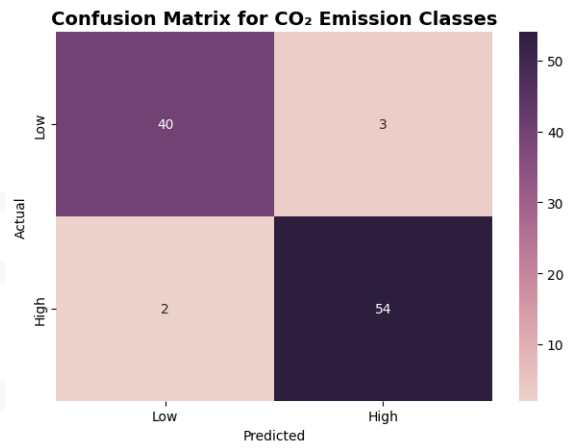


Figure 4.6: Confusion Matrix for CO₂ Emissions

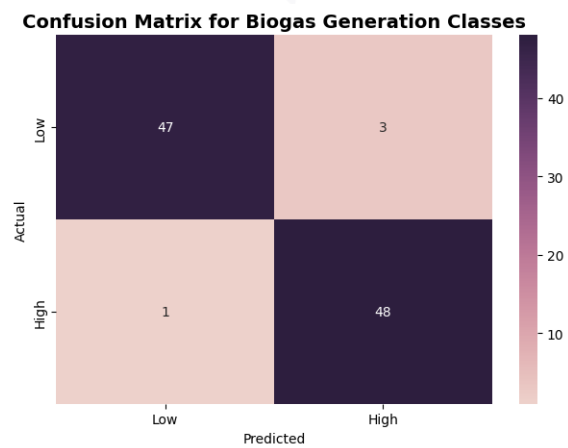


Figure 4.7: Confusion Matrix for Biogas Generation

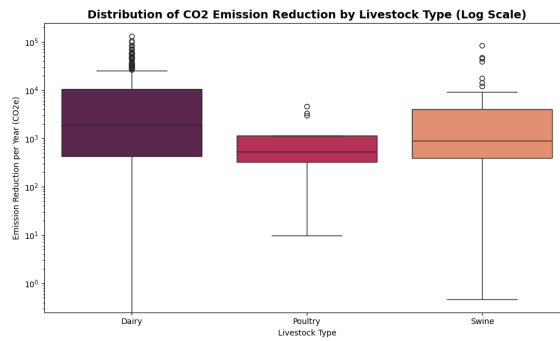


Figure 4.8: Performance by Livestock Type

high, the models depend on the data of past projects, shifts in technology might diminish predictive stability over the time. Even then, these results bridge the research gap by providing evidence based predictions for energy generation and CO₂ emissions.

4.3 Comparative Analysis of Livestock Types

To understand the differences in digester performance, they were analyzed based on livestock types.

Swine digesters: Illustrates most variety

Poultry digesters: Shows high efficiency for CO₂ emission reduction

Dairy digesters: Display the highest median electricity outputs

As displayed in Figure 4.8

Different livestock have different impacts on the digester performance. This is due to the specific properties of manure. Dairy digesters are dominant among others because of their consistent biogas yield and higher fiber content. The high nitrogen content in swine manure can either increase methane production or lead to the instability of the process depending upon features such as soil conditions, forms of nitrogen, availability of carbon substrates etc.

4.4 Regional & Spatial Patters

Digester adoption is highest in California, Wisconsin and New York. As can be seen in the Figure 4.9, displays the outcome of the code and a map of the US for the identification of different states.

Electricity Generation

As far as the regions are concerned, **Midwest** demonstrates the highest concentration of digesters with a broad range electricity generation. **West** has a low number of projects than others but some of them are the highest producers of electricity. **Northeast and**

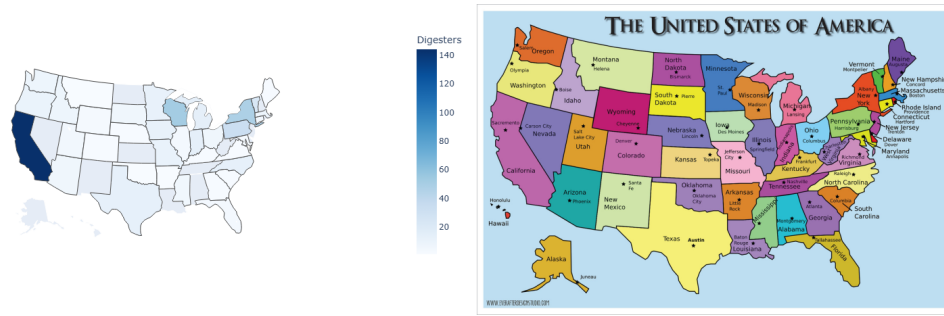


Figure 4.9: State Level Adoption of Digesters. Source: [Studio, 2020]

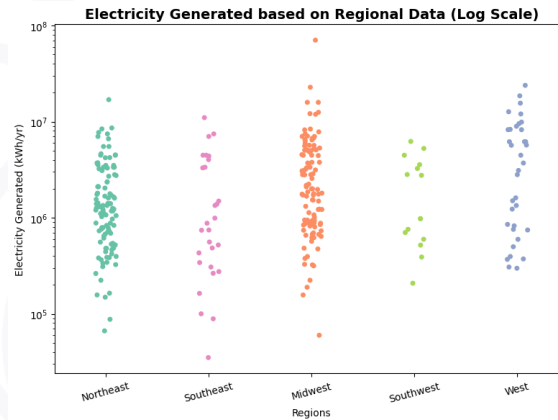


Figure 4.10: Electricity Generated by Region

Southeast have a few extremes and lastly, **Southwest** also has few projects but are of a wide range. A scatterplot Figure 4.10 showing regional electricity generation.

CO₂ Emission Reduction

Regionally, **Midwest** seems to be the backbone of emission reductions. **West** has few projects but they play an important role in CO₂ reduction. Overall, Northeast and Southeast seem to be moderate performers but are still important whereas **Southwest** is least impactful. Another scatterplot Figure 4.11 to show regional outcomes. Another than this, an html map shows the highest performers for the CO₂ emission reduction.

Geography is a very important aspect in determining if the digester operations are a success or not. California and Wisconsin are identified as significant for energy generation programs and large dairy businesses. This indicates that further digester development plans should be region specific and supported by both technical and policy factors.

4.5 Model Interpretability

For CO₂ Emissions, SHAP values indicate that the project age and waste volume are the biggest drivers of CO₂ emission reduction. Depicted in Figure 4.12. Rest of the features only have a minor influence.

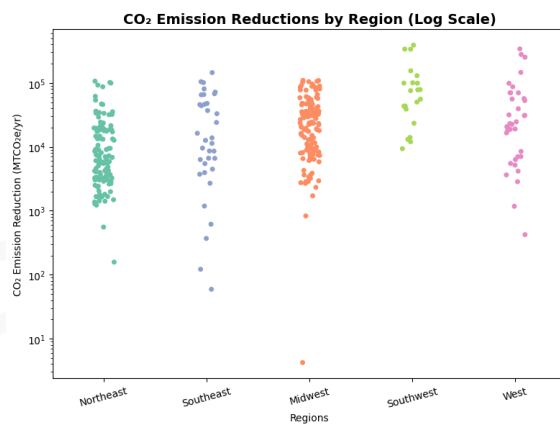


Figure 4.11: Emission Reduction based on Region

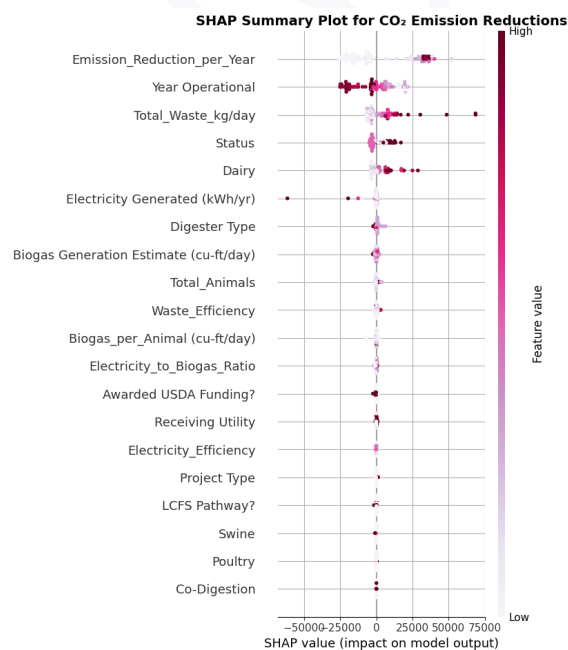


Figure 4.12: SHAP Analysis for Emission Reduction

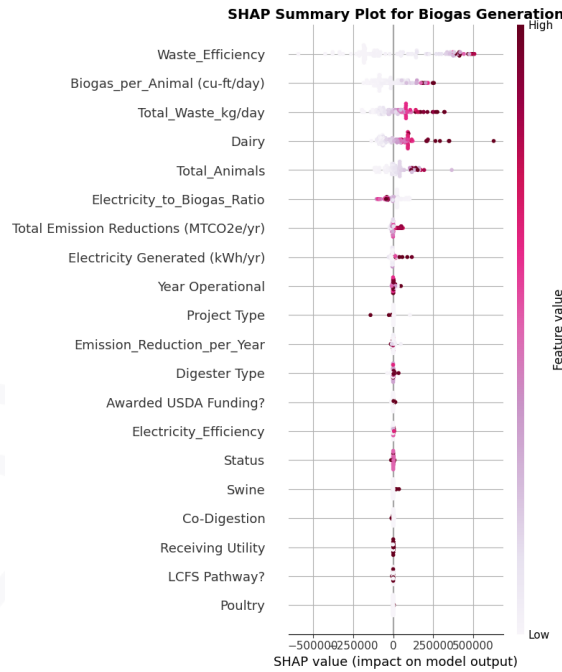


Figure 4.13: SHAP Analysis for Biogas Generation

Likewise, for biogas generation, waste efficiency, dairy animals and waste volume plays a key role in affecting the predictions. As shown in Figure 4.13. Interpretability results using SHAP reveal that large scale projects are not always the efficient ones and this is why increasing the size of digesters does not necessarily guarantee a bigger return. SHAP uses real farm data unlike ADM1's complex biochemical inputs and helps in bridging gaps between research and its application.

4.6 Overall Insights

- Although dairy based digesters dominate the US digesters, swine and poultry also demonstrate unexplored potential.
- Machine learning models are way more accurate in predicting energy generation and emission reduction than the traditional methods.
- Geography and policy both have a great influence in digester deployment and efficiency.
- Analysis through SHAP highlights the important features and helps in understanding the digesters on other levels.

These results fill the gaps in the research by incorporating regional comparisons, overcoming the bias towards dairy digesters and utilizing ML to predict and explain digesters,

their working and the outputs.

By identifying high performing digester projects, using ML predictions, investors can lower the risk and boost funding confidence. By lowering emissions and generating energy renewable energy, strategic planning could have in achieving climate targets.



Chapter 5

Conclusion

The aim of this study was to analyze the efficiency of livestock based anaerobic digesters in the US with a focus on using machine learning to predict energy production and CO₂ emission reduction. This work has produced some quantitative results and a thorough understanding of a digester's role in sustainable energy and agricultural systems by applying the Random Forest and XGBoost algorithms to a dataset of 491 digester projects. Although there were several swine and poultry digesters, the dairy digesters outperformed the others in terms of output. For CO₂ reduction and electricity generation, XGBoost predicted R², Random Forest predicted the classification reports, both demonstrating remarkable predictive power. Among the states with the greatest digester adoption rates were California, Wisconsin, and New York. States in the Midwest produced more energy and reduced emissions more than those in the Southwest. By employing real world data instead of mechanistic models, this study provides a method that strikes a good mix between geography, interpretability, and accurate predictions. By adding updated datasets, with projects after 2019, the study could be further developed in the future. Furthermore, integrating machine learning models with real time operational data can improve dynamic predictions and facilitate cross border comparisons that aid in comprehending the function of digesters in energy generation. The study shows that machine learning can be an effective technique for assessing and predicting the performance of digesters that rely on livestock. New information about the locations and ways digesters contribute to energy production and emission reduction has been made possible by the integration of geographic mapping, livestock analysis, and predictive modeling. The results show that, although they depend on several circumstances, digesters play a significant role in sustainability. A significant step towards a more robust and low carbon agriculture sector would be the expansion of anaerobic digesters as a component of the US renewable energy system.

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